

ESP32 + Programming bits

The diagram illustrates the wiring for an ESP32-WROOM-32 module. Key components and connections include:

- Power and Grounding:** A 10µF capacitor (C1) and a 0.1µF capacitor (C3) are connected to the VCC and GND pins. The module's GND pin is connected to a common ground point labeled 1.15.38.
- Programming:** A 4-pin header is connected to VCC, GND, RX, and TX pins.
- Buttons:** A 4-pin header is connected to EN, BTN_01, BTN_02, and BTN_03 pins.
- Other Connections:** The module's V_BAT_READ, A_PIN, SPEAKER, IR_CLK_EN, and BTN_13 pins are connected to various points on the board.
- Internal Flash:** The module's internal flash memory is connected to the JTAG pins (1, 2, 3, 4) via resistors R3, R4, R5, and R6.
- GPIOs:** The module's GPIO pins (IO0 to IO23) are connected to various points on the board, including the JTAG pins and the internal flash.

[illegible][illegible]

Expansion

Diagram illustrating the expansion header (J3) on the back of the Wiimote. The header is a 6-pin connector with the following labels and connections:

- VCC (Pin 1)
- SCL (Pin 2)
- EXT_Sense (Pin 3)
- NC (Pin 4)
- SDA (Pin 5)
- GND (Pin 6)

The expansion header is labeled J3. The pin numbers 3, 1, 6, 4, 2, and 5 are highlighted in a red box, indicating the sequence of pins for the expansion module.



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Strapping pins

The diagrams illustrate the wiring for four strapping pins:

- BOOT_SW:** A 10kΩ resistor (R1) connects VCC to pin 2. Pin 1 is connected to the BOOT pin of the TLS3315. Pin 3 is connected to GND.
- GP105:** A 10kΩ resistor (R10) connects VCC to pin 2. Pin 1 is connected to the GP105 pin.
- EN_SW:** A 10kΩ resistor (R2) connects VCC to pin 2. Pin 1 is connected to the EN pin of the EN_SW component. Pin 3 is connected to GND.
- GP102:** A 10kΩ resistor (R9) connects GND to pin 2. Pin 1 is connected to the GP102 pin.

IR Camera

The diagram illustrates the connection between an IR camera module and a PiXart WII camera module. The IR camera module is represented by a yellow box with a red border, containing a schematic symbol for a camera and the text "Tri-State". It has three pins: VDD (pin 4), OUT (pin 3), and GND (pin 2). The PiXart WII module is represented by a yellow box with a red border, labeled "PiXart_WII (PiXart_WII)" and "J2". It has a 7-pin header with the following labels: PAD, RESET, IR_CLK, SDA, SCL, NC, GND2, GND1, and 3.3V. The connections are as follows: VCC is connected to VDD (pin 4) and GND (pin 2). IR_CLK_EN is connected to IR_CLK (pin 5). OUT (pin 3) is connected to SDA (pin 6). GND (pin 2) is connected to GND1 (pin 1) and GND2 (pin 3). The 3.3V pin is connected to GND (pin 1). The RESET pin (pin 7) is connected to GND (pin 1). The PAD pin (pin 2) is connected to GND (pin 1). The SCL pin (pin 4) is connected to GND (pin 1). The NC pin (pin 6) is connected to GND (pin 1).

BATTERY BOOST

I2C Pullups

The image contains two circuit diagrams illustrating I2C pullups. The top diagram shows a green line labeled 'SDA' connected to a yellow rectangular resistor labeled 'R12' with a value of '4.7kΩ'. The other end of the resistor is connected to a red arrow labeled 'VCC'. The bottom diagram shows a green line labeled 'SCL' connected to a yellow rectangular resistor labeled 'R14' with a value of '4.7kΩ'. The other end of the resistor is connected to a red arrow labeled 'VCC'.

Aiming for ~2mA

Rumble Motor

TEST SPEAKER WITH RAW SPEAKER PIN
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